SCIENCEUTSAV®
Science is awesome

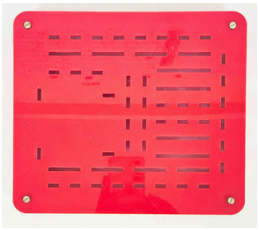
I'm a ROBO SCIENTIST

Light Sensing and Automatic Lighting Systems

Session

8

STRUCTURE AND FASTENERS OF MICROCONTROLLERS

1


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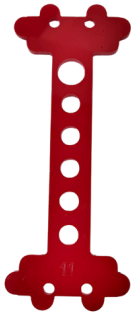


1

Spacers

2

L Clamp

3

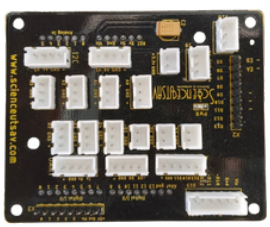
Spacer Screws

4

Nuts

5

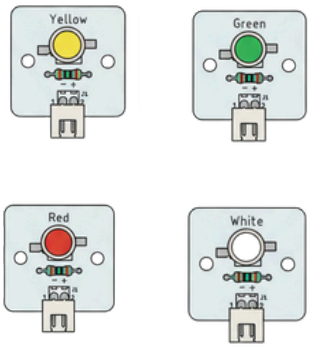
Screw

6

Arduino With Shield

7

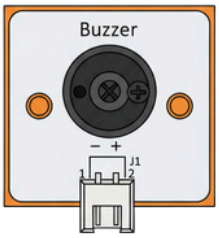
Arduino cable

ELECTRONIC COMPONENTS



1 Watt LED PCB

1 Watt LED is a high-brightness light-emitting diode that produces much stronger light compared to normal small LEDs.



Buzzer

A buzzer is an electronic sound-producing device that generates audible tones or beeps when powered. It is commonly used in alarms, notifications, indicators, security systems, and electronic projects to provide sound-based alerts and feedback.

Push Button

A push button is a simple mechanical switch that controls an electrical circuit when pressed. It is widely used in electronic devices, control panels, automation systems, and DIY projects to provide user input and trigger specific actions.

LDR Sensor (Light Dependent Resistor)

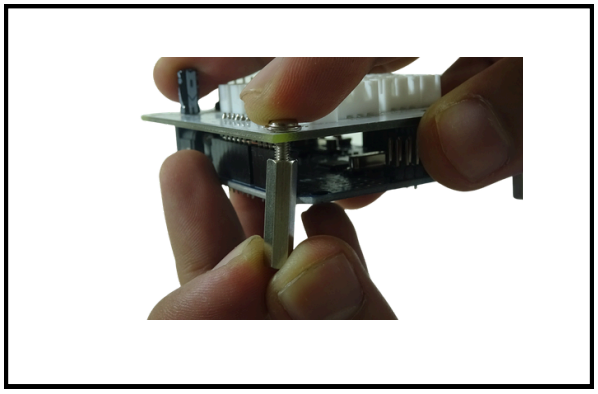
An LDR sensor is a light-sensitive electronic component that changes its resistance according to the intensity of light falling on it. It is commonly used in automatic lighting systems, light detection circuits, security devices, and electronic projects to sense and respond to changes in ambient light.

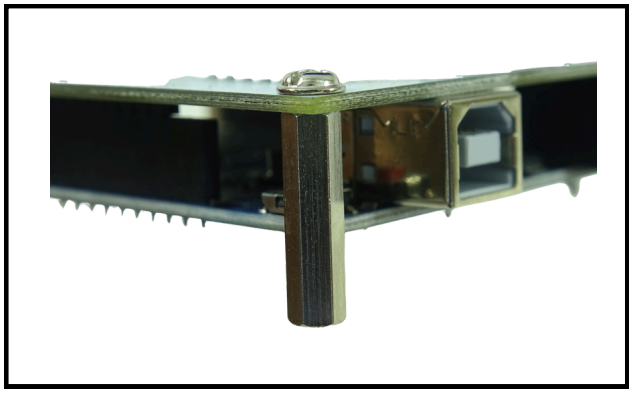
2 PIN RMC

Used to connect the 1-watt LEDs and the buzzer.

Procedure :

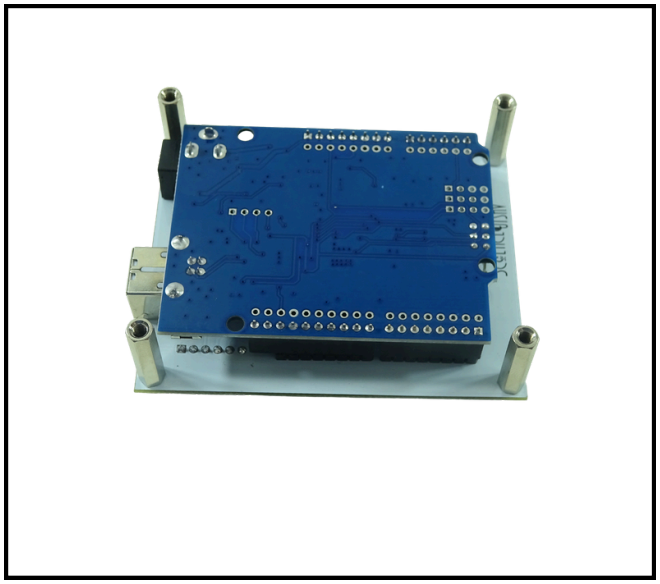
Step 1 : Fit the spacer to Arduino Shield as shown below using Spacer screw

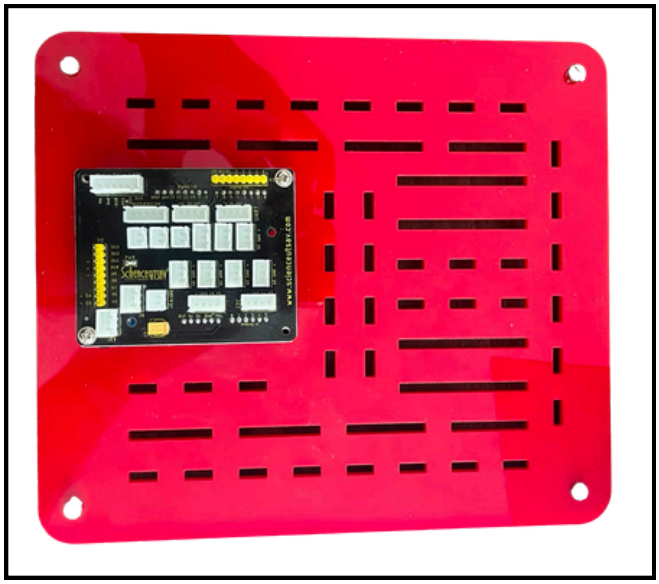




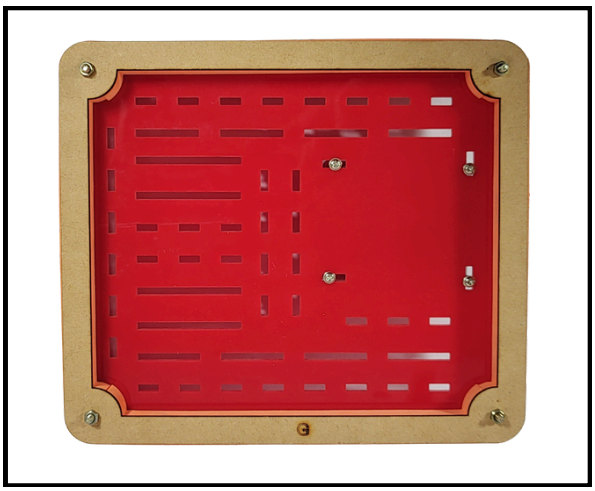
Step 2 : Repeat the step 1 for all 4 corners of Arduino Shield

Step 3 : Place the Arduino Shield to the base board as shown below



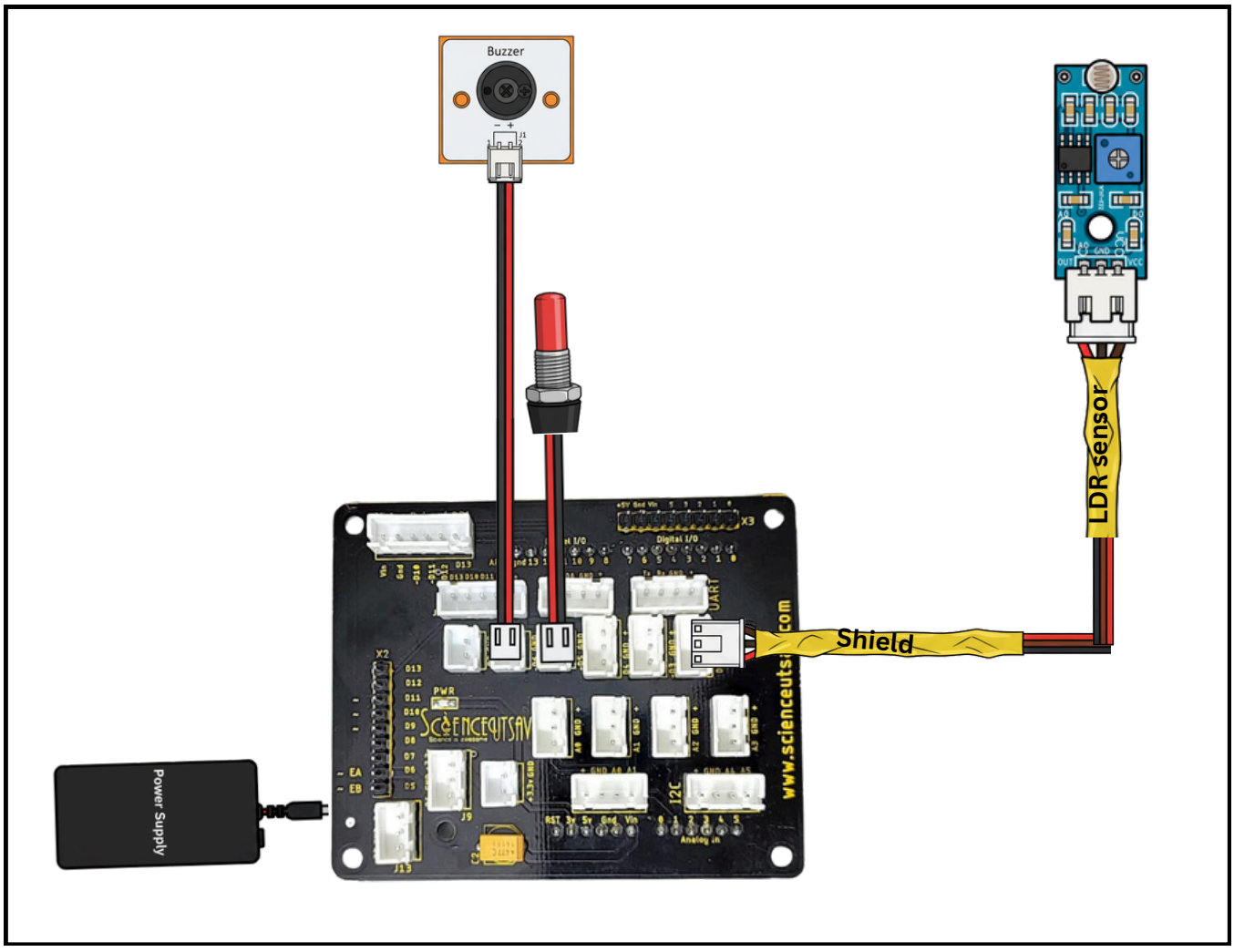


Step 4 : Fix it with the spacer screw from the bottom as shown below



BEGINNER PROJECT

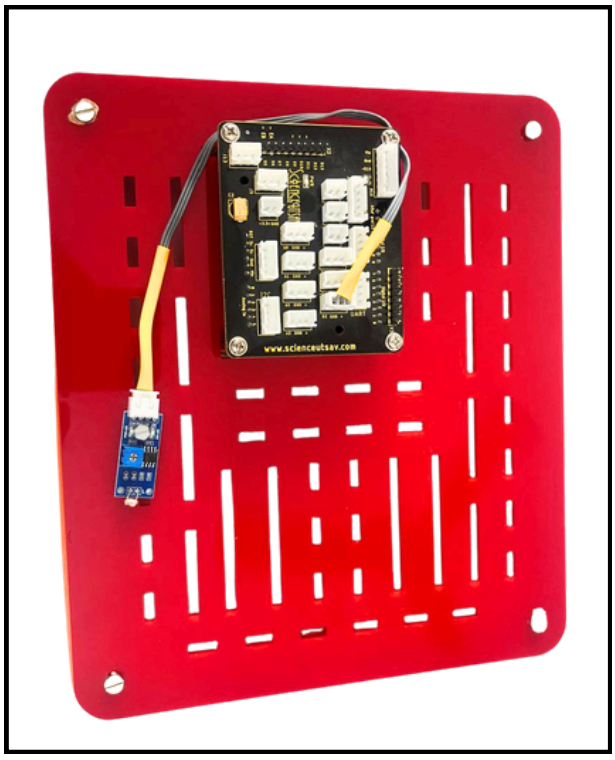
The circuit diagram of the project is given below



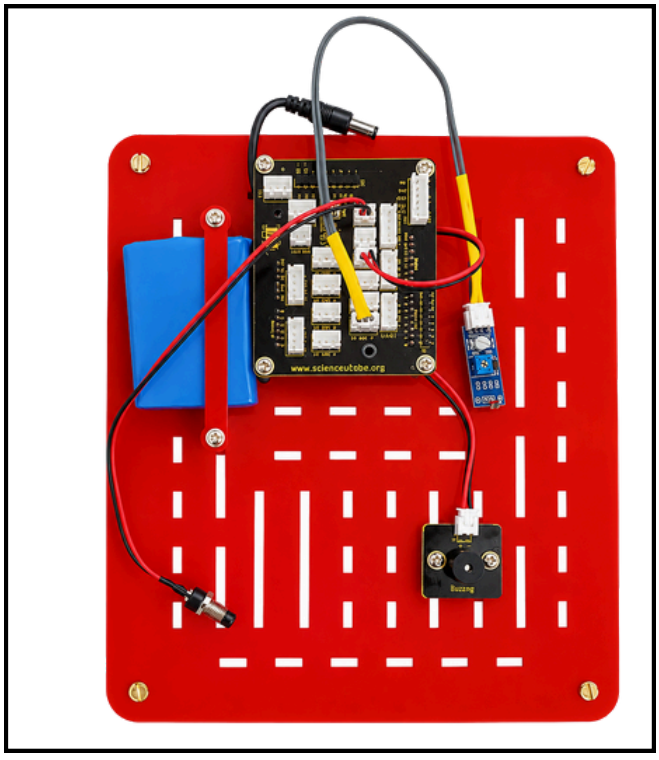
CONNECTIONS:

1. LDR SENSOR - D2
2. PUSH BUTTON - D5
3. BUZZER - D6

Step 1- Connect the LDR sensor to the board

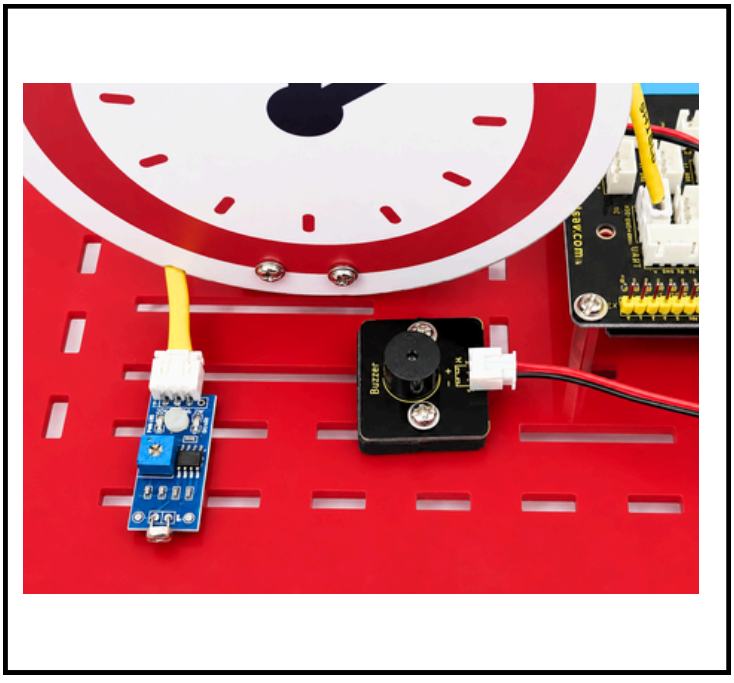


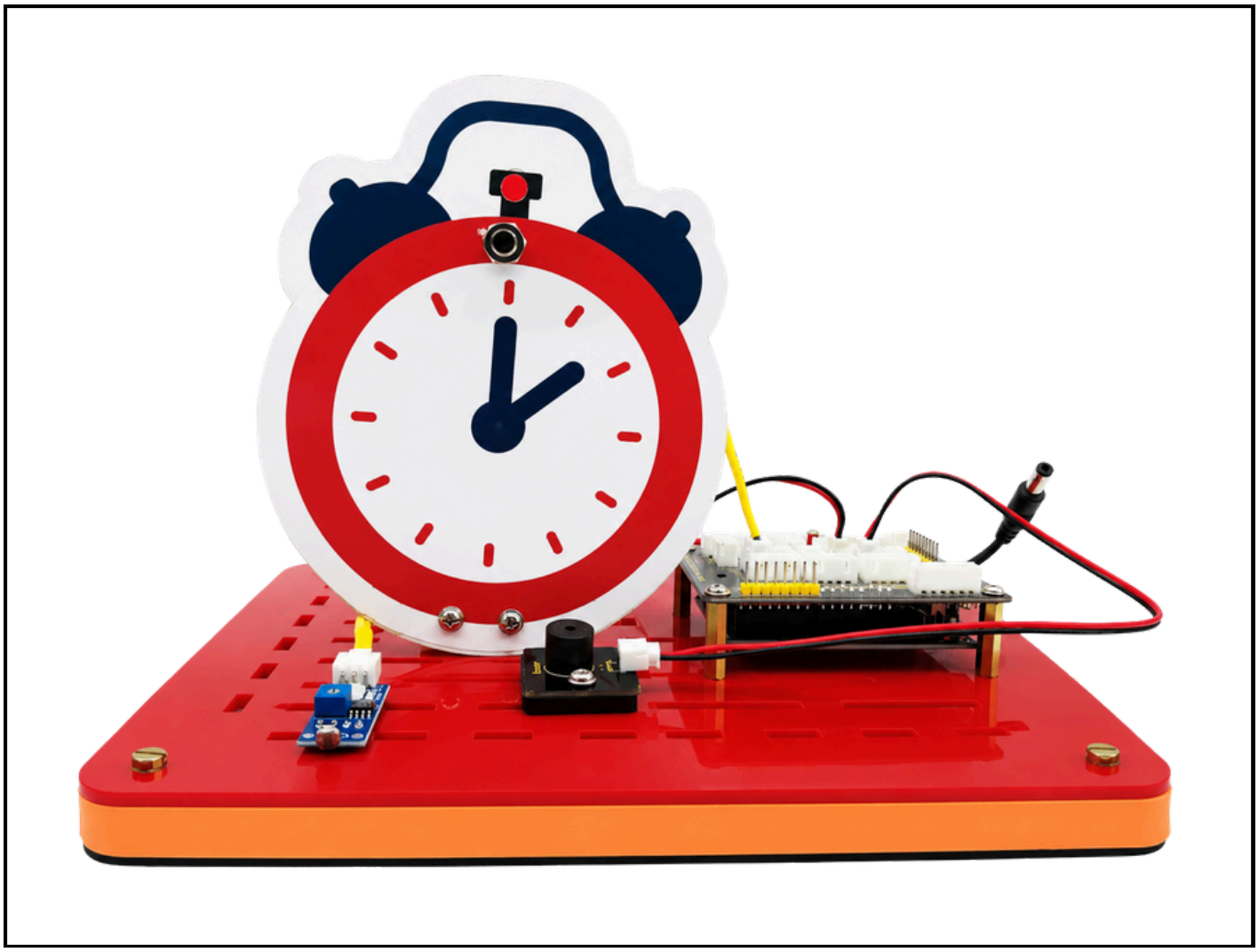
Step 2- Connect the buzzer and the push button as shown



Step 3- Fix the template to the stand and insert the push button

Step 4- Fix the buzzer infront of the alarm





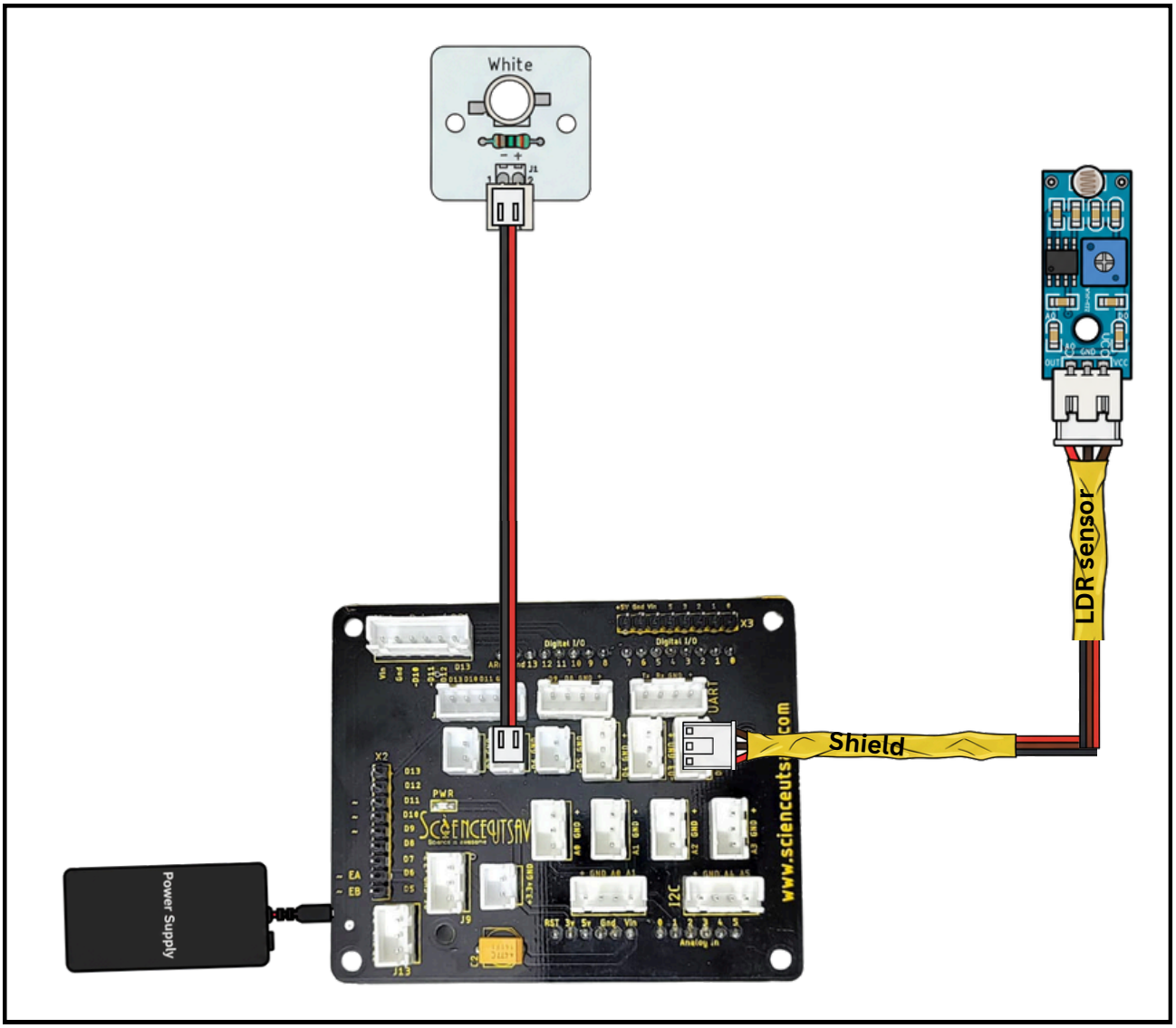
Your project is now ready for exploration!

Code

```
/*
ScienceUtsav Arduino Code - Session 6 to 10
Corrected for 3-pin DIGITAL LDR module.
LDR uses digitalRead(), not analogRead().
*/

const int ldrSensor = 2;      // 3-pin LDR digital OUT
const int pushSwitch = 5;
const int buzzer = 6;

const int DARK_STATE = LOW;   // If LDR works opposite, change LOW to HIGH
bool alarmOff = false;

void setup() {
  pinMode(ldrSensor, INPUT);
  pinMode(pushSwitch, INPUT_PULLUP);
  pinMode(buzzer, OUTPUT);
}

void loop() {
  bool darkDetected = digitalRead(ldrSensor) == DARK_STATE;

  if (digitalRead(pushSwitch) == LOW) {
    alarmOff = true;
    noTone(buzzer);
    delay(300);
  }

  if (darkDetected && !alarmOff) {
    tone(buzzer, 1200);
  } else {
    noTone(buzzer);
  }

  if (!darkDetected) alarmOff = false;
}
```

BEGINNER PROJECT

The circuit diagram of the project is given below

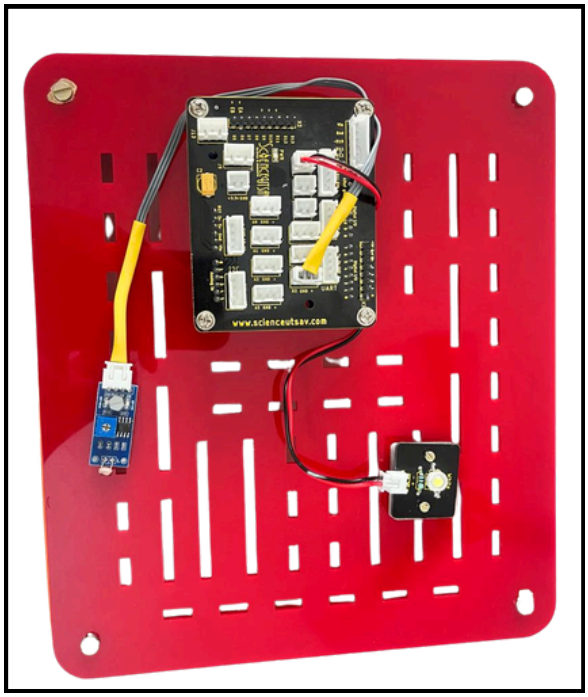


CONNECTIONS:

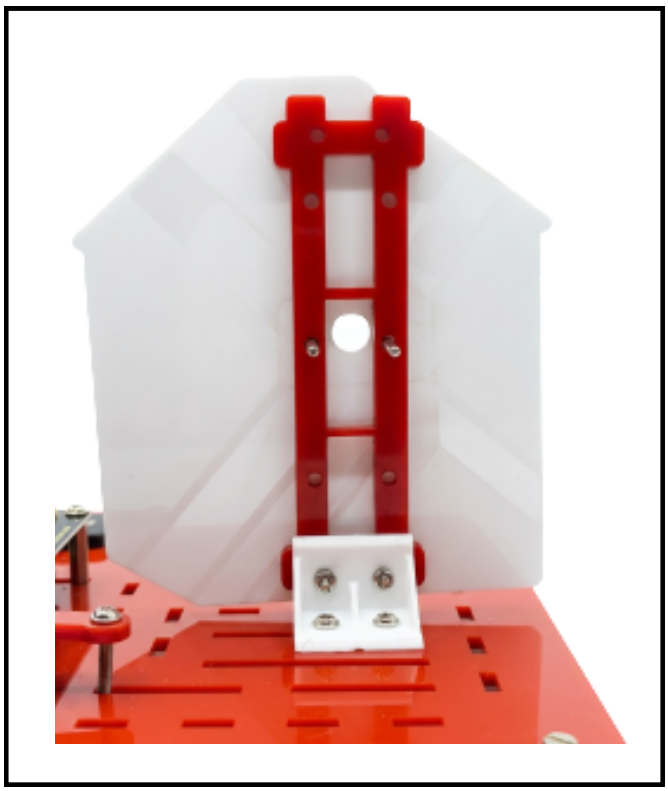
1. LDR SENSOR - D2
2. 1W LED - D5

Step 1- Connect the LDR sensor to the board

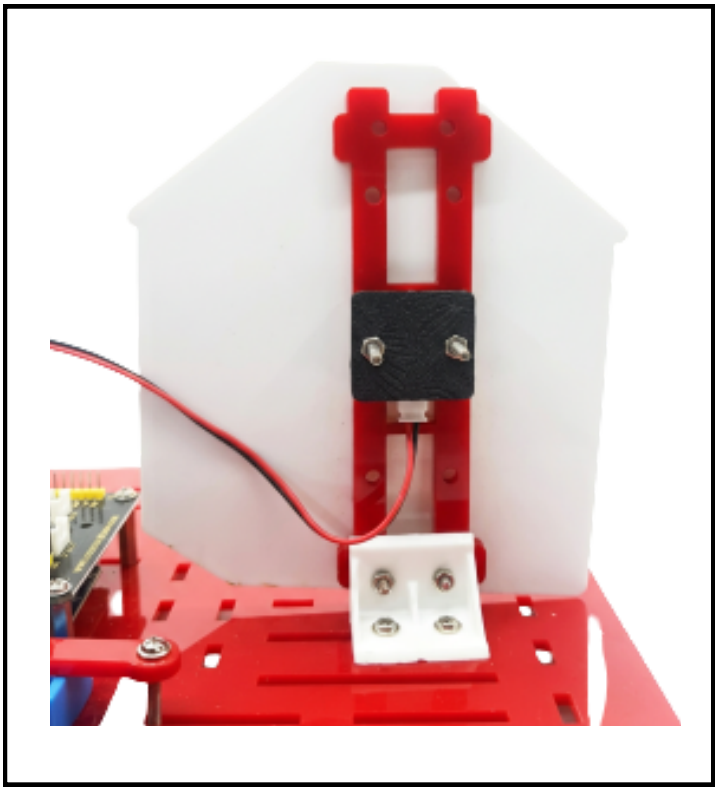
Step 2- Connect the LED as shown

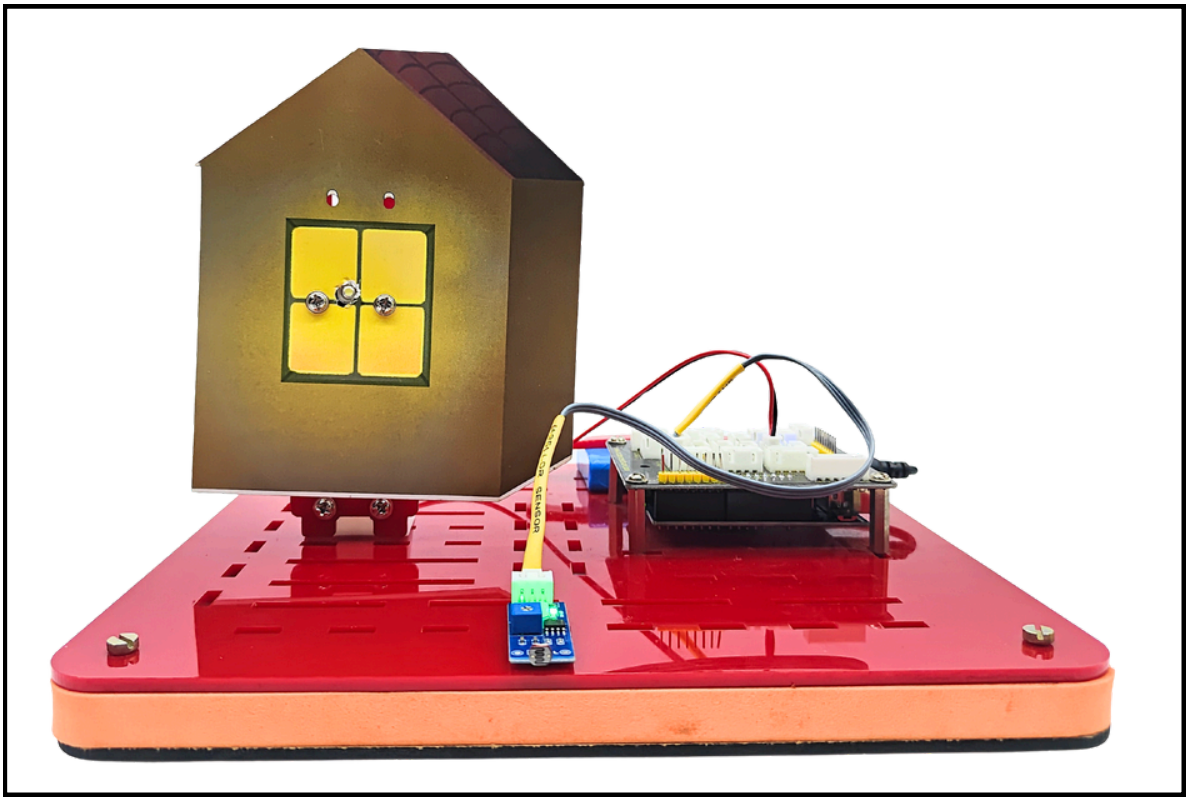


Step 3- Attach the stand and fix the template to it.



Step 4- Attach the LED using the bolts as show





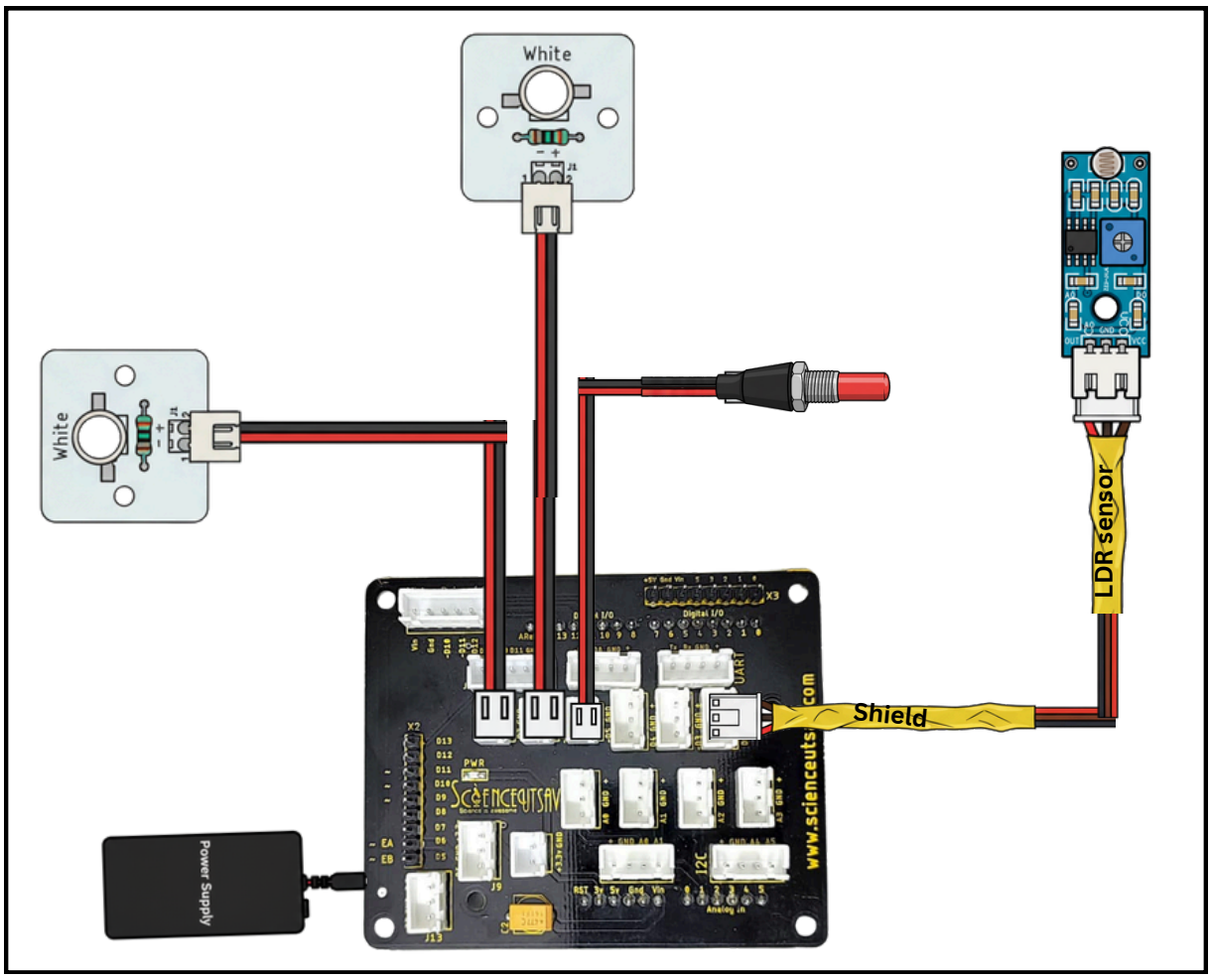
The window is ready to light up!

Code

```
/*
ScienceUtsav Arduino Code - Session 6 to 10
Corrected for 3-pin DIGITAL LDR module.
LDR uses digitalRead(), not analogRead().
*/

const int ldrSensor = 2;
const int led = 6;
const int DARK_STATE = HIGH;

void setup() {
  pinMode(ldrSensor, INPUT);
  pinMode(led, OUTPUT);
}

void loop() {
  bool darkDetected = digitalRead(ldrSensor) == DARK_STATE;
  digitalWrite(led, darkDetected ? HIGH : LOW);
}
```

INTERMEDIATE PROJECT

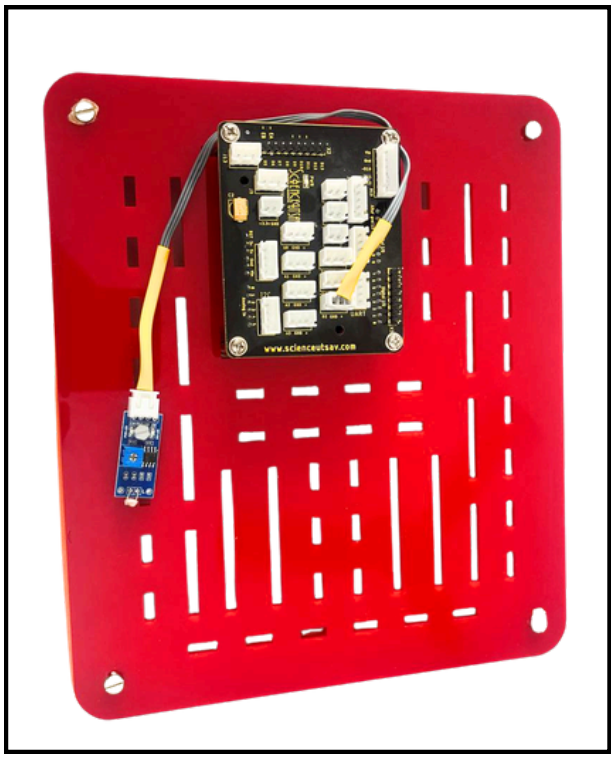
The circuit diagram of the project is given below



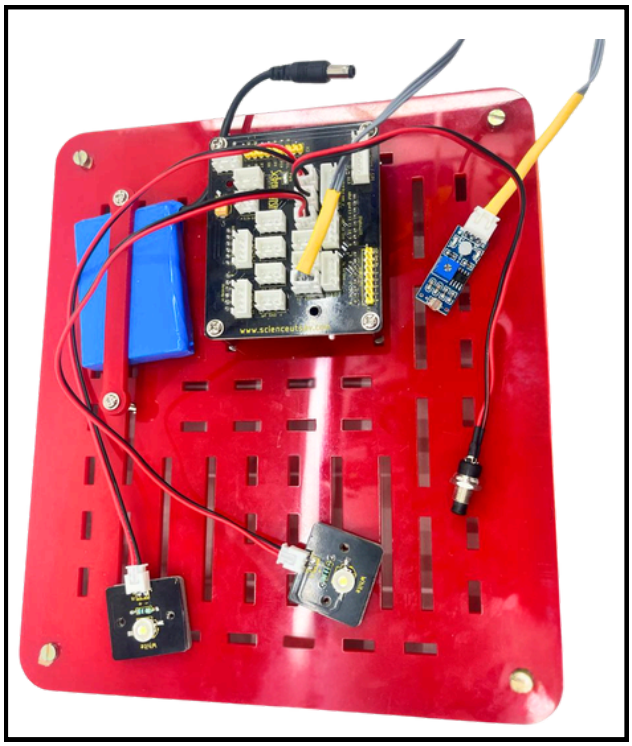
CONNECTIONS:

1. LDR SENSOR - D2
2. PUSH BUTTON - D5
3. 1W LEDs - D6, D7

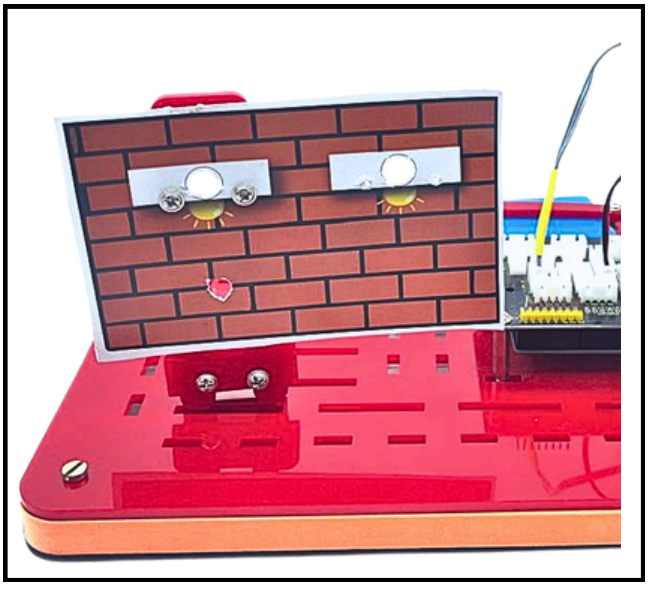
Step 1- Connect the LDR sensor to the board



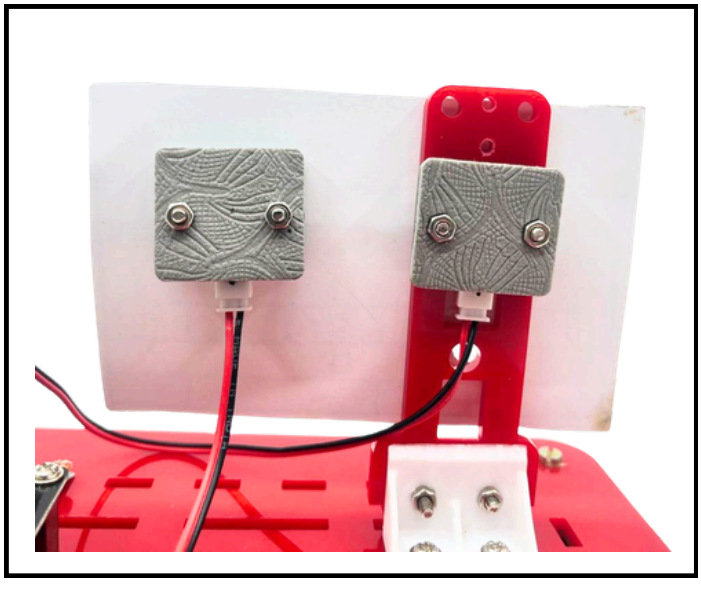
Step 2- Connect the LEDs and the push button as shown



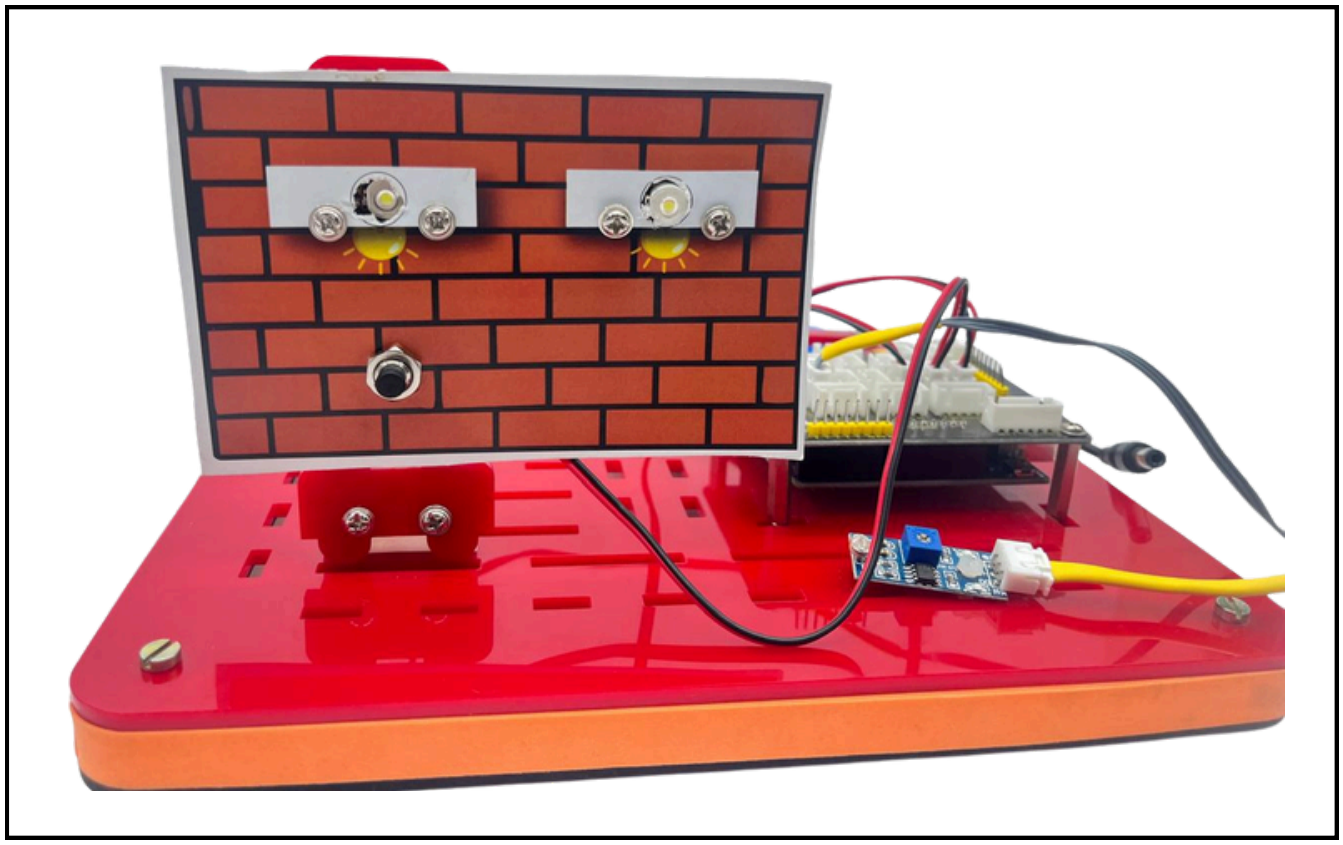
Step 3- Fix the stand to the template as shown and attach it to the board



Step 4- Attach both the LEDs to the template



Step 5- Insert the push button into the template



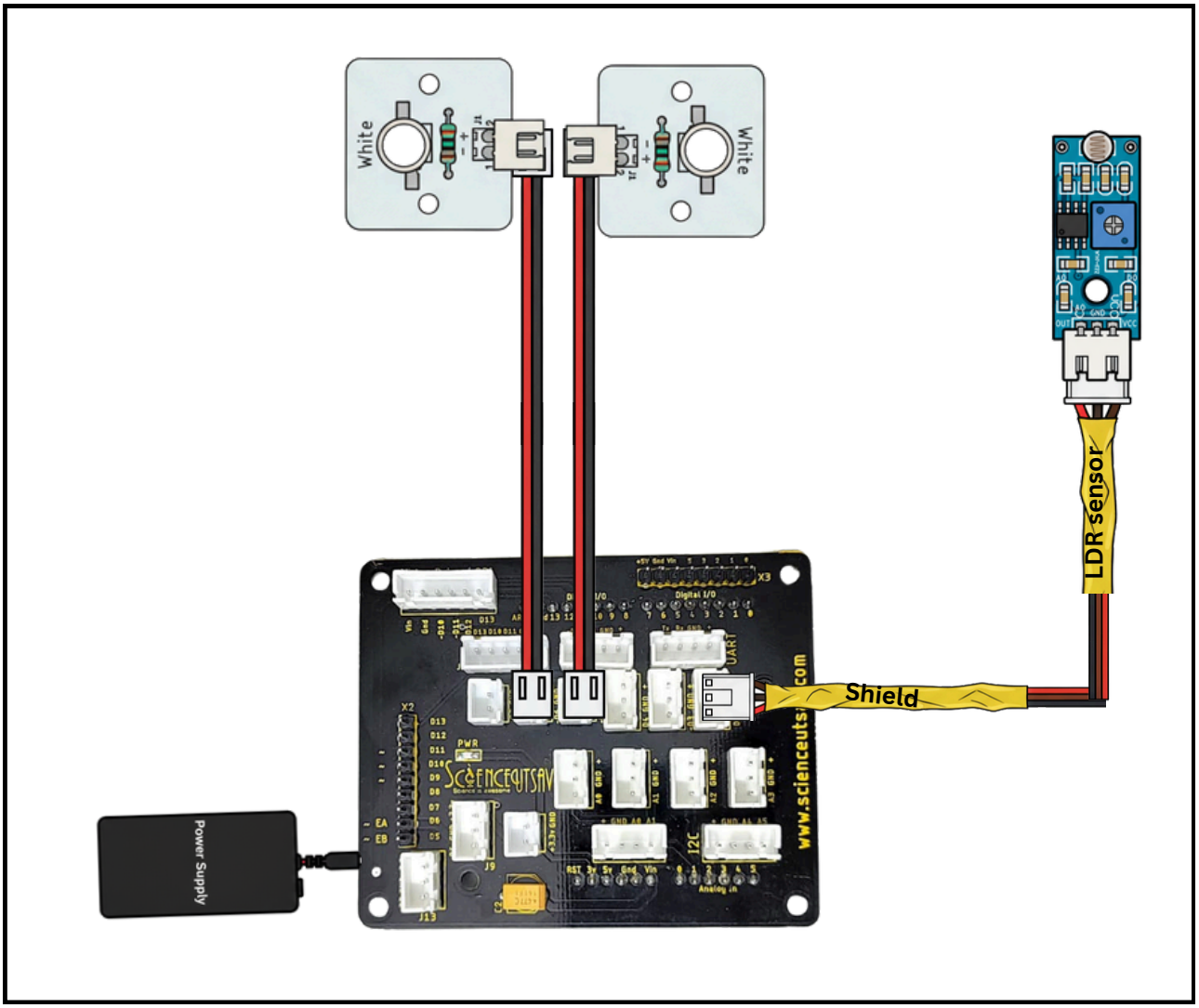
Light up the wall!

Code

```
/*
ScienceUtsav Arduino Code
Digital LDR + Push Switch OFF control
*/

const int ldrSensor = 2;
const int switchPin = 5;

const int led1 = 6;
const int led2 = 7;

const int DARK_STATE = HIGH;

void setup() {
  pinMode(ldrSensor, INPUT);
  pinMode(switchPin, INPUT_PULLUP);

  pinMode(led1, OUTPUT);
  pinMode(led2, OUTPUT);
}

void loop() {
  bool darkDetected = digitalRead(ldrSensor) == DARK_STATE;
  bool switchPressed = digitalRead(switchPin) == LOW;

  if (switchPressed) {
    // Switch pressed means LEDs OFF
    digitalWrite(led1, LOW);
    digitalWrite(led2, LOW);
  }
  else {
    // LEDs ON only when dark
    digitalWrite(led1, darkDetected ? HIGH : LOW);
    digitalWrite(led2, darkDetected ? HIGH : LOW);
  }
}
```

INTERMEDIATE PROJECT

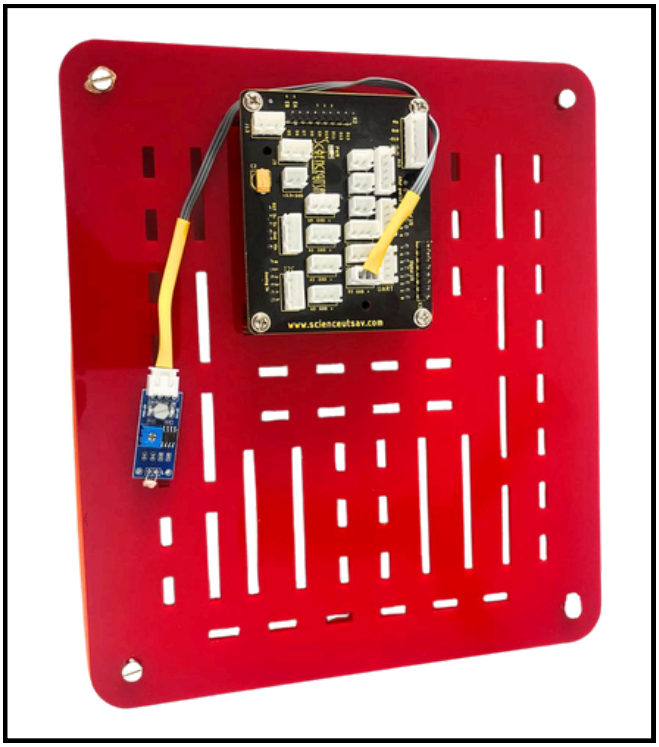
The circuit diagram of the project is given below



CONNECTIONS:

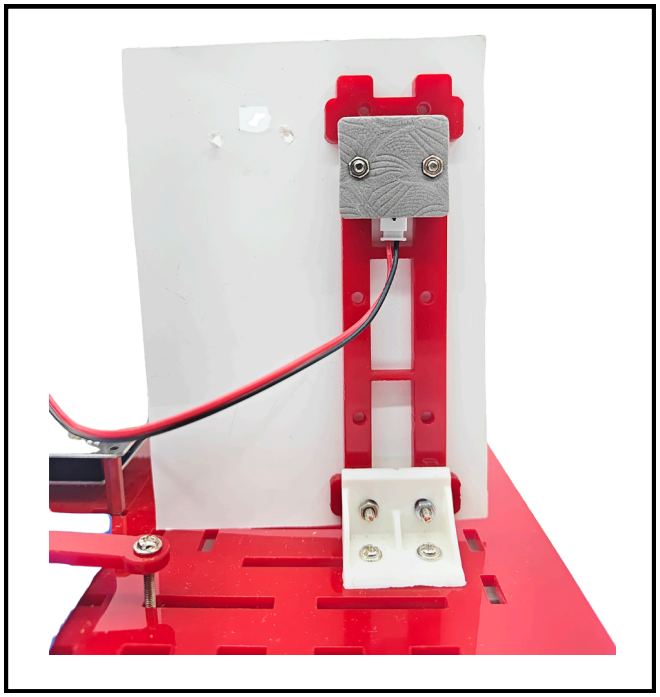
1. LDR SENSOR - D2
2. 1W LEDs - D5, D6

Step 1- Connect the LDR sensor to the board

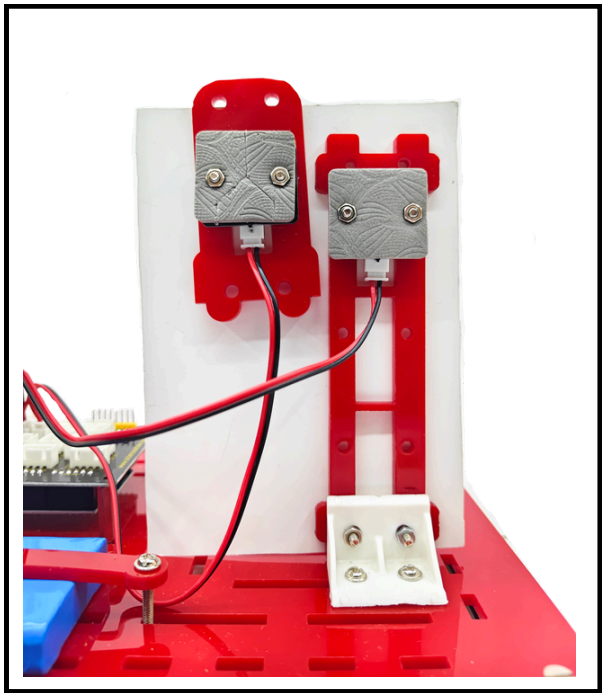


Step 2- Attach the stand along with the template

Step 3- Attach the LED as shown



Step 4- Attach the other stand along with the LED



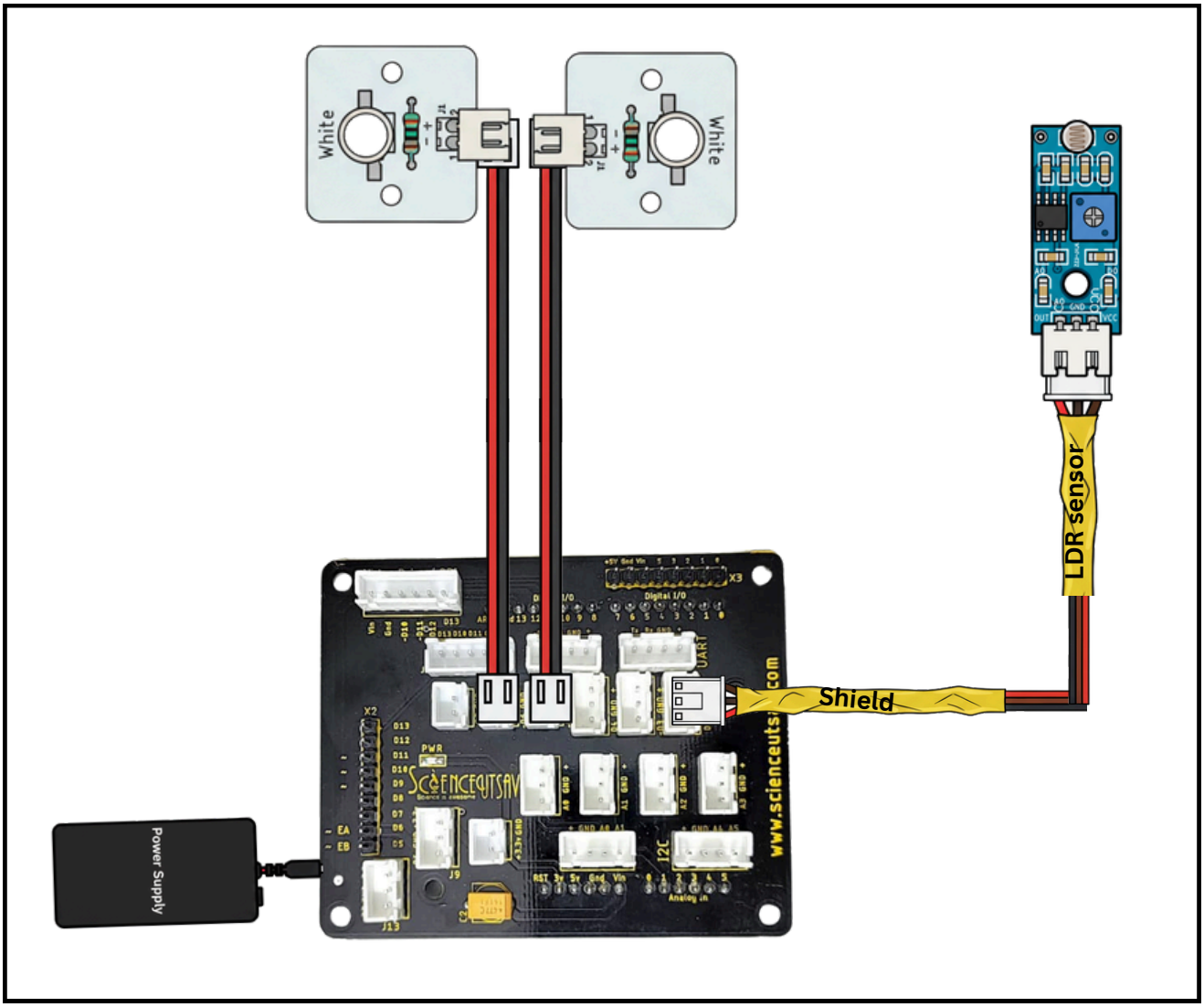
Light up the street!

Code

```
/*
ScienceUtsav Arduino Code - Session 6 to 10
Corrected for 3-pin DIGITAL LDR module.
LDR uses digitalRead(), not analogRead().
*/

const int ldrSensor = 2;
const int led1 = 5;
const int led2 = 6;

const int DARK_STATE = HIGH;

void setup() {
  pinMode(ldrSensor, INPUT);
  pinMode(led1, OUTPUT);
  pinMode(led2, OUTPUT);

}

void loop() {
  bool darkDetected = digitalRead(ldrSensor) == DARK_STATE;

  digitalWrite(led1, darkDetected ? HIGH : LOW);
  digitalWrite(led2, darkDetected ? HIGH : LOW);


}
```

ADVANCE PROJECT

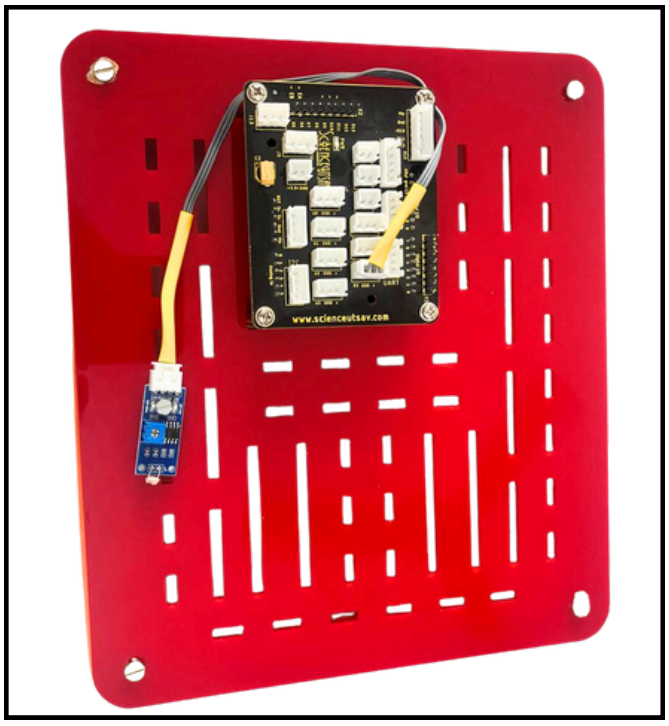
The circuit diagram of the project is given below



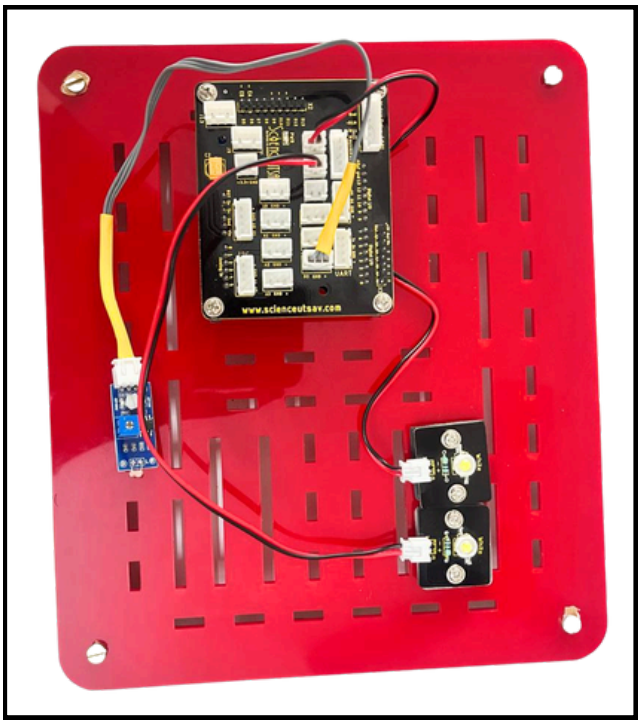
CONNECTIONS:

1. LDR SENSOR- D2
2. 1W LEDs- D5, D6

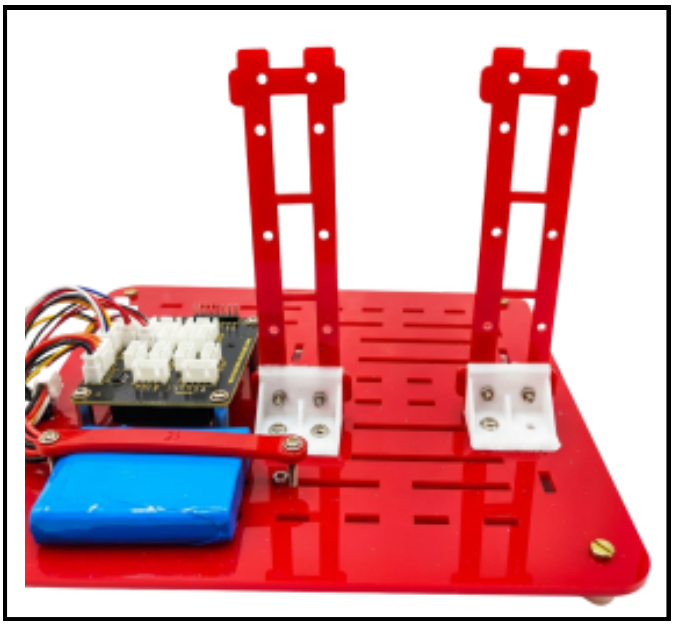
Step 1- Connect the LDR sensor to the board



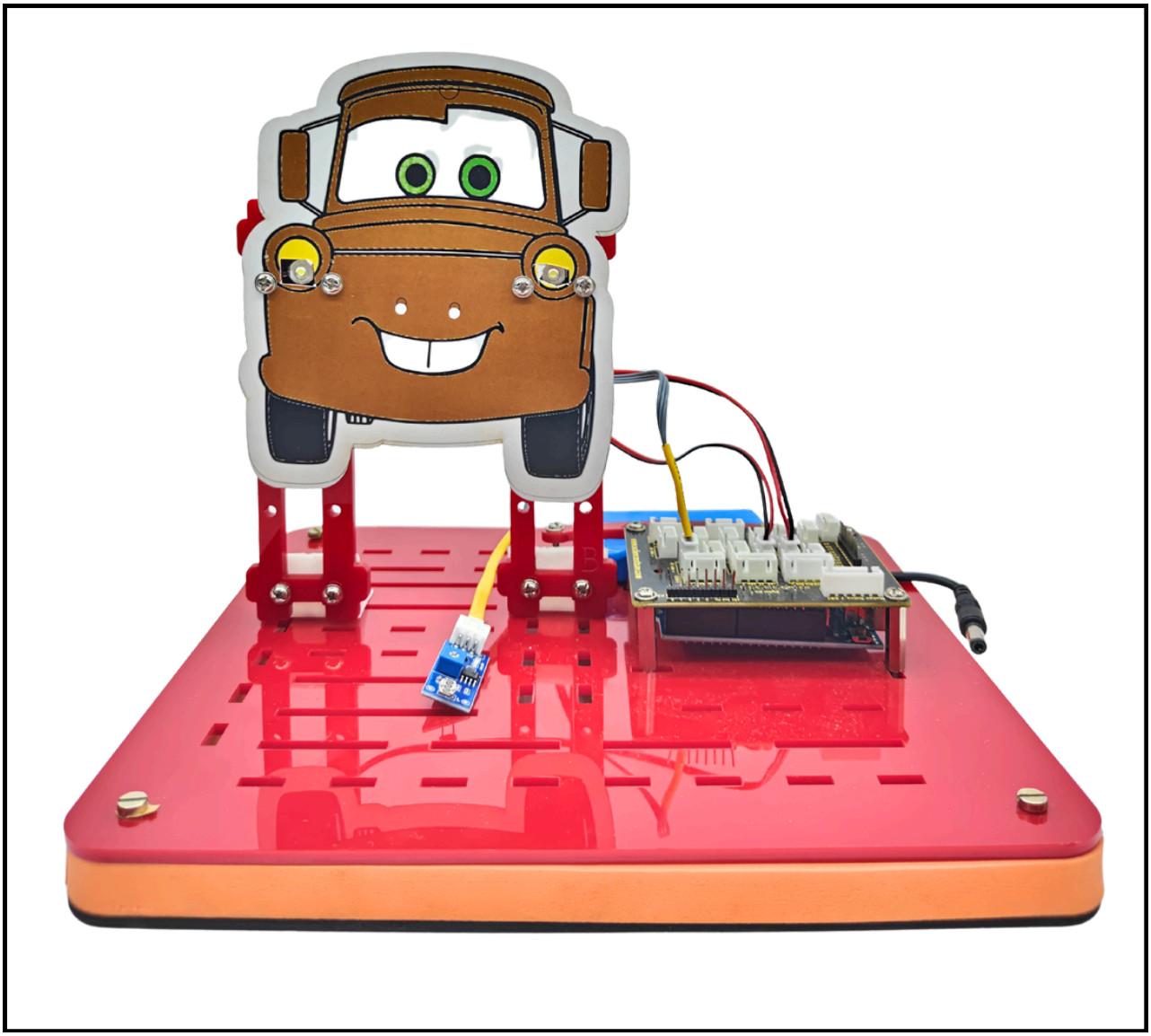
Step 2- Attach the two LEDs as shown



Step 3- Fix the stands onto the board



Step 4- Fix the template along with the LEDs



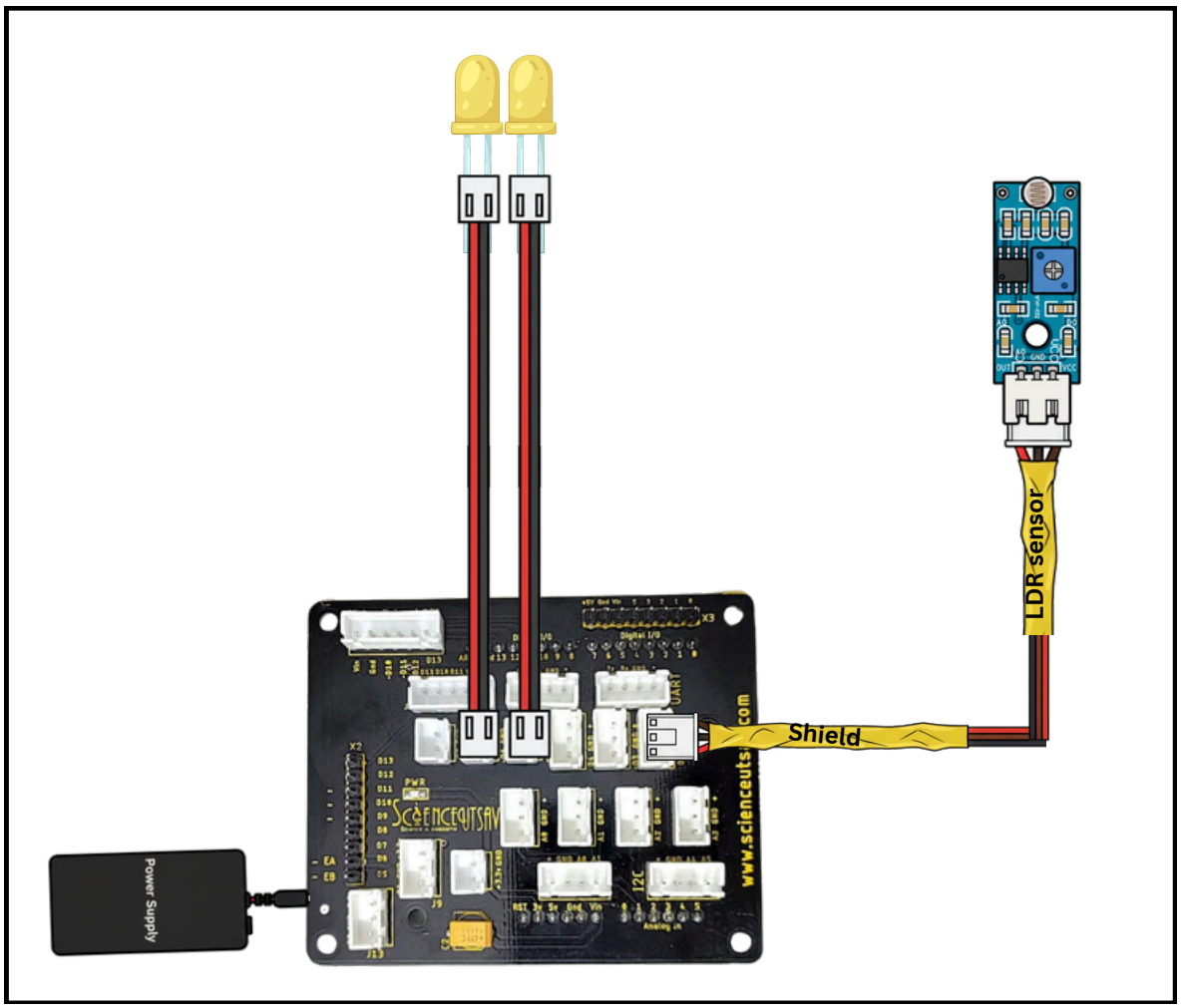
Watch the headlights turn on on their own!

Code

```
/*
ScienceUtsav Arduino Code - Session 6 to 10
Corrected for 3-pin DIGITAL LDR module.
LDR uses digitalRead(), not analogRead().
*/

const int ldrSensor = 2;
const int led = 6;
const int rgbRed = 10;
const int rgbGreen = 11;
const int rgbBlue = 12;
const int DARK_STATE = LOW;

void setup() {
  pinMode(ldrSensor, INPUT);
  pinMode(led, OUTPUT);
  pinMode(rgbRed, OUTPUT);
  pinMode(rgbGreen, OUTPUT);
  pinMode(rgbBlue, OUTPUT);
}

void loop() {
  bool darkDetected = digitalRead(ldrSensor) == DARK_STATE;

  if (darkDetected) {
    digitalWrite(led, HIGH);
    digitalWrite(rgbRed, HIGH);
    digitalWrite(rgbGreen, HIGH);
    digitalWrite(rgbBlue, HIGH);
  } else {
    digitalWrite(led, LOW);
    digitalWrite(rgbRed, LOW);
    digitalWrite(rgbGreen, LOW);
    digitalWrite(rgbBlue, LOW);
  }
}
```

ADVANCE PROJECT

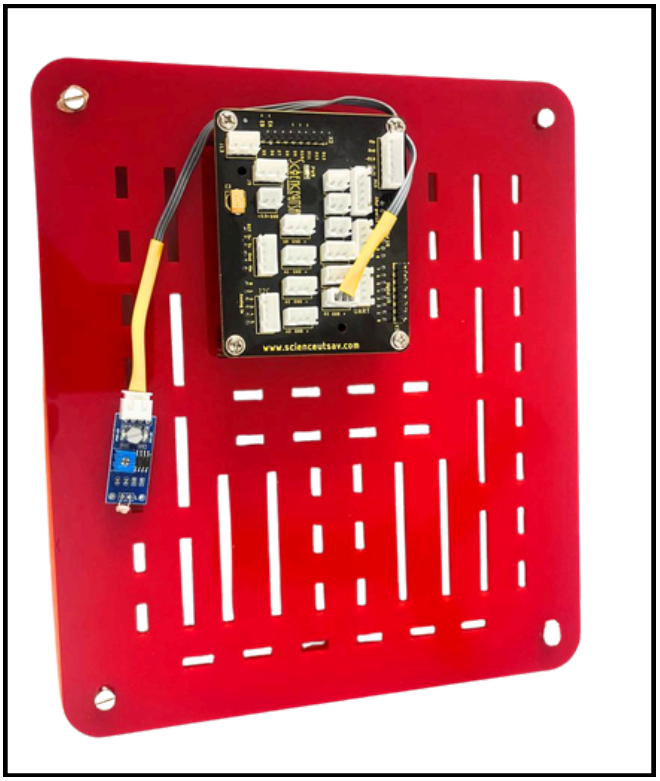
The circuit diagram of the project is given below



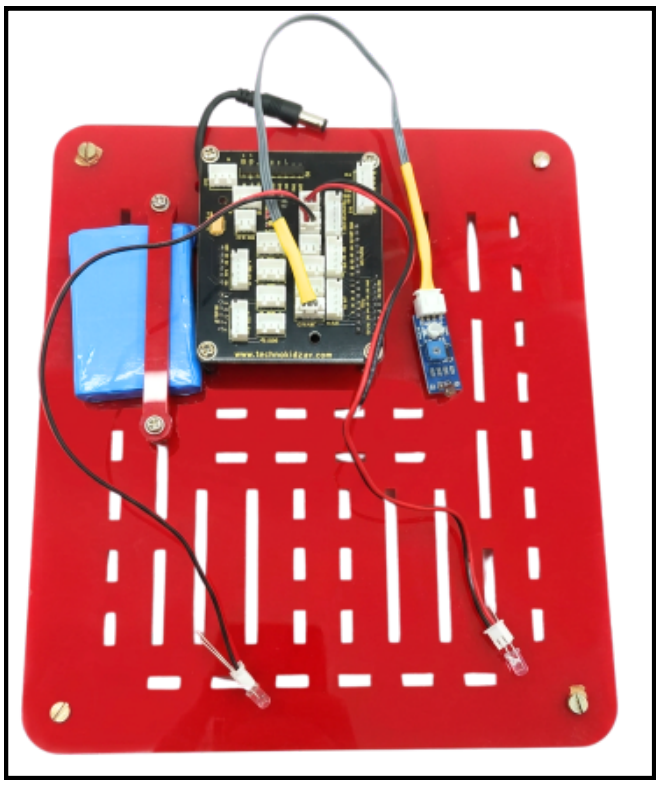
CONNECTIONS:

1. LDR SENSOR - D2
2. DIP LEDs - D5, D6

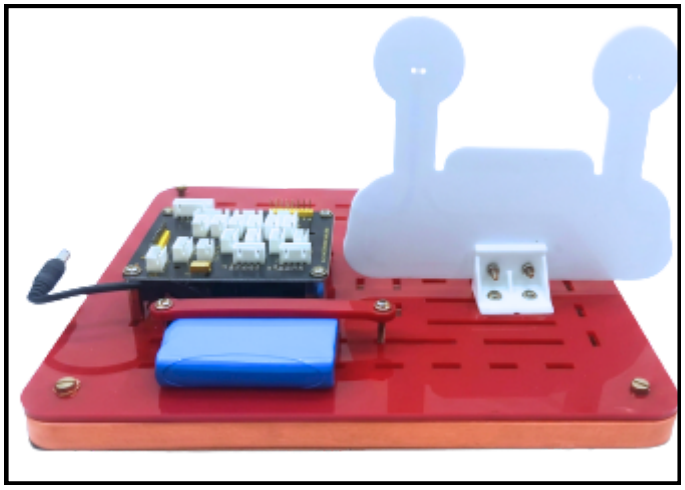
Step 1- Connect the LDR sensor to the board



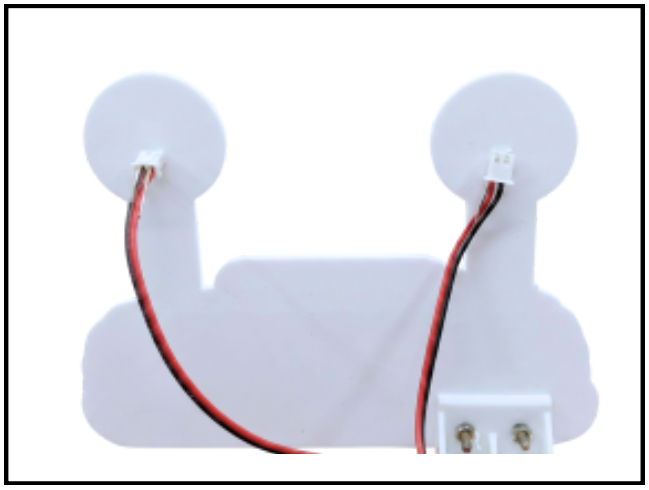
Step 2- Connect the dip LEDs as shown



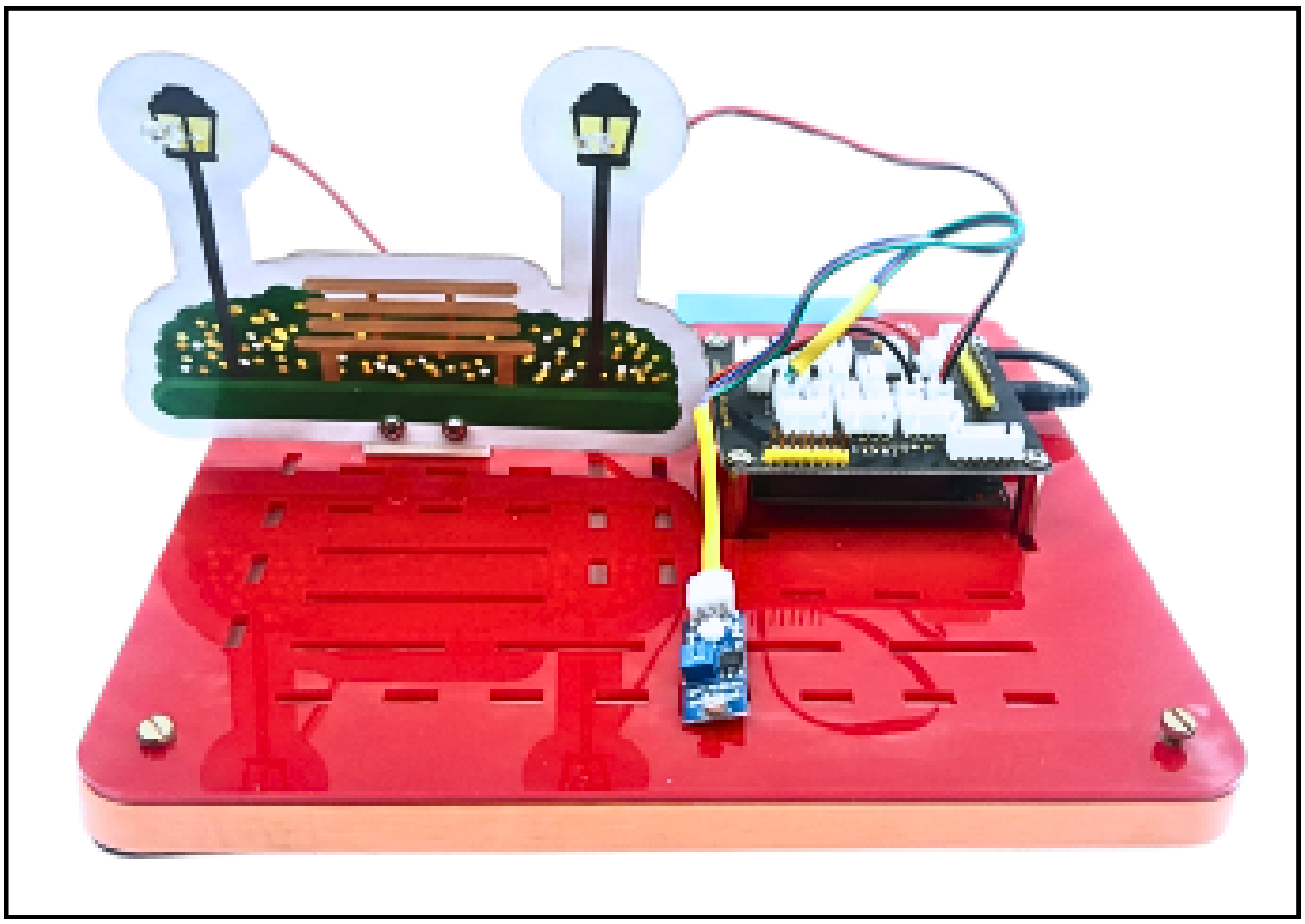
Step 3- Fix the stand onto the board



Step 4- Insert the LEDs through the front and the wires through the back



Remember- The long leg of the LED goes to the red wire and the shorter one goes to the black wire



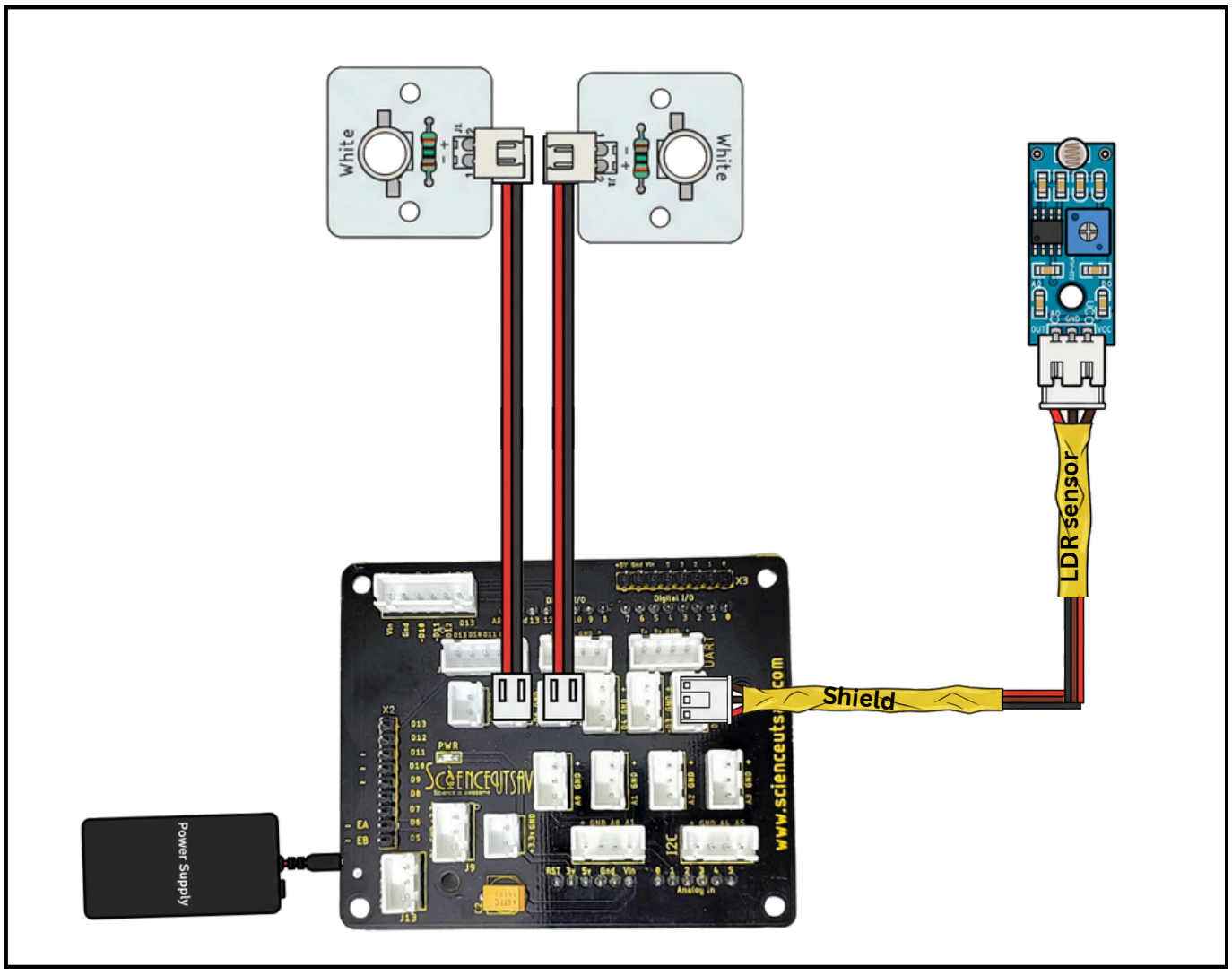
Enjoy the glow of the streetlamps!

Code

```
/*
ScienceUtsav Arduino Code - Session 6 to 10
Corrected for 3-pin DIGITAL LDR module.
LDR uses digitalRead(), not analogRead().
*/

const int ldrSensor = 2;
const int led1 = 5;
const int led2 = 6;
const int DARK_STATE = HIGH;

void setup() {
  pinMode(ldrSensor, INPUT);
  pinMode(led1, OUTPUT);
  pinMode(led2, OUTPUT);
}

void loop() {
  bool darkDetected = digitalRead(ldrSensor) == DARK_STATE;
  digitalWrite(led1, darkDetected ? HIGH : LOW);
  digitalWrite(led2, darkDetected ? HIGH : LOW);
}
```

EXPERT PROJECT

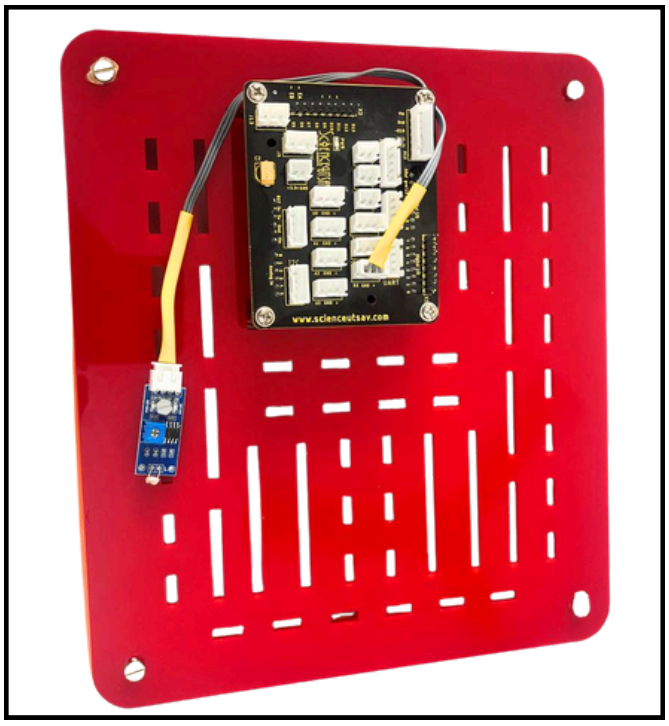
The circuit diagram of the project is given below



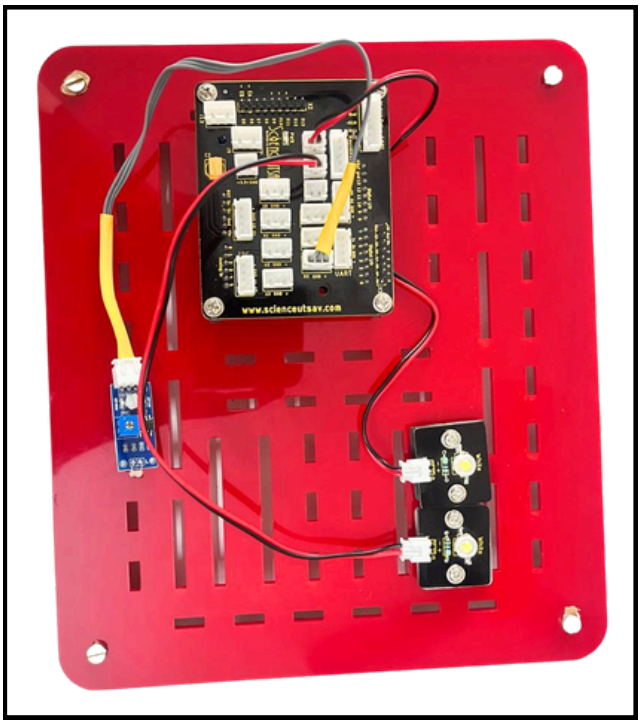
CONNECTIONS:

1. LDR SENSOR - D2
2. 1W LEDs - D5, D6

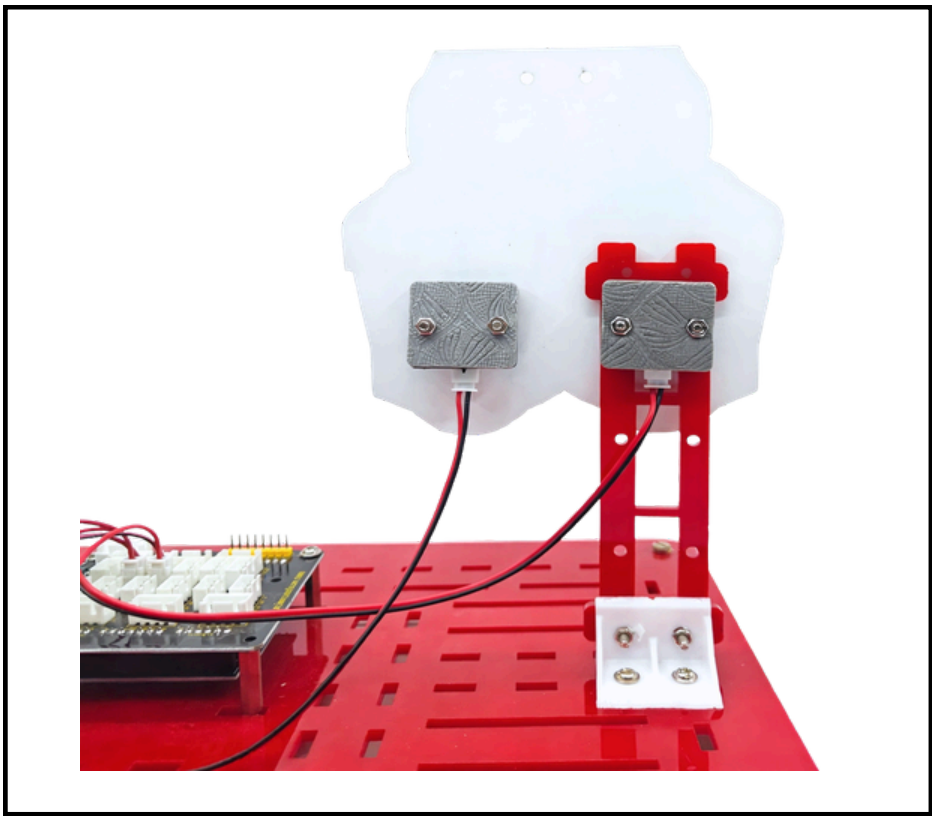
Step 1- Connect the LDR sensor to the board

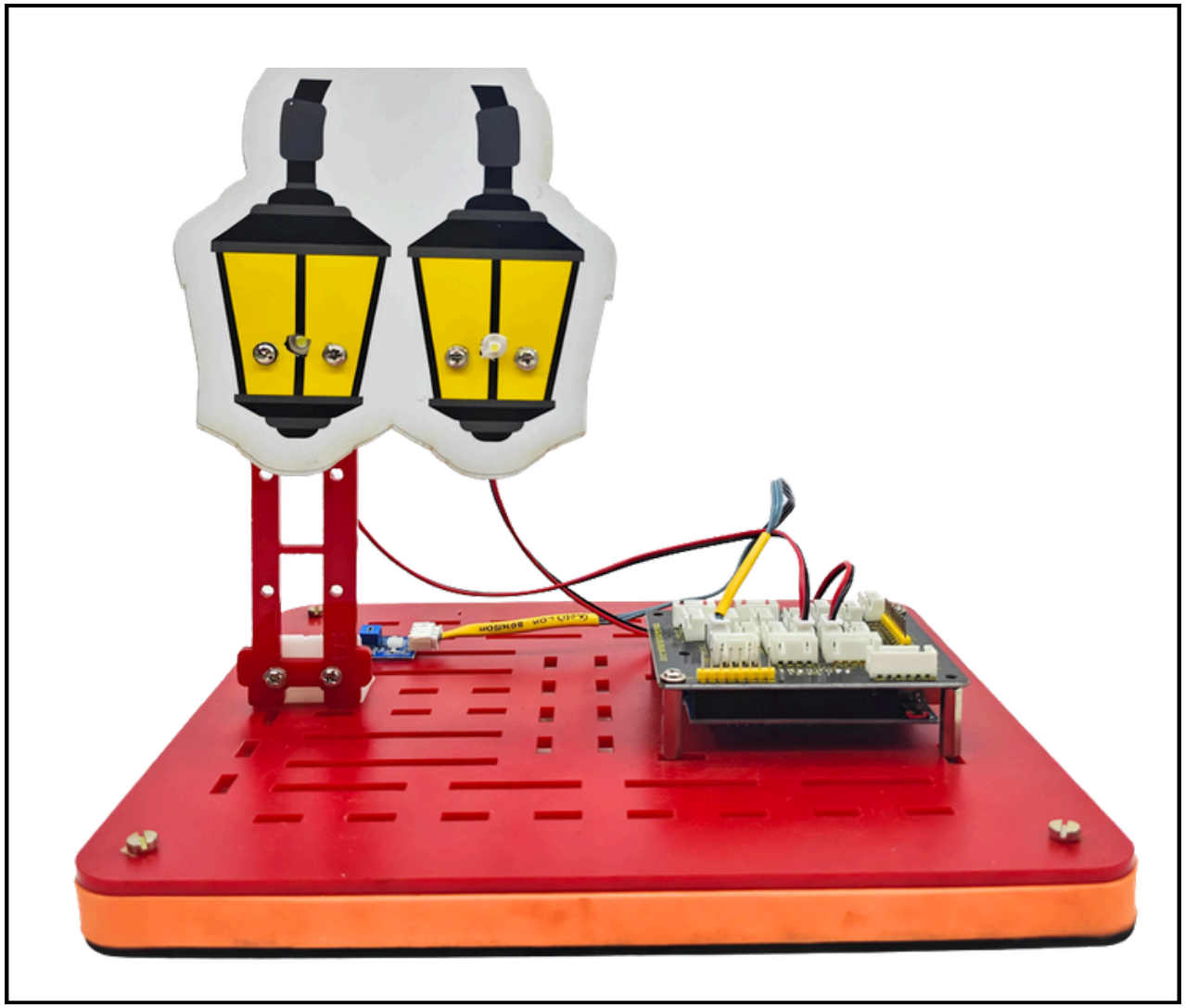


Step 2- Attach the two LEDs as shown



Step 3- Attach the template along with the LEDs to the stand





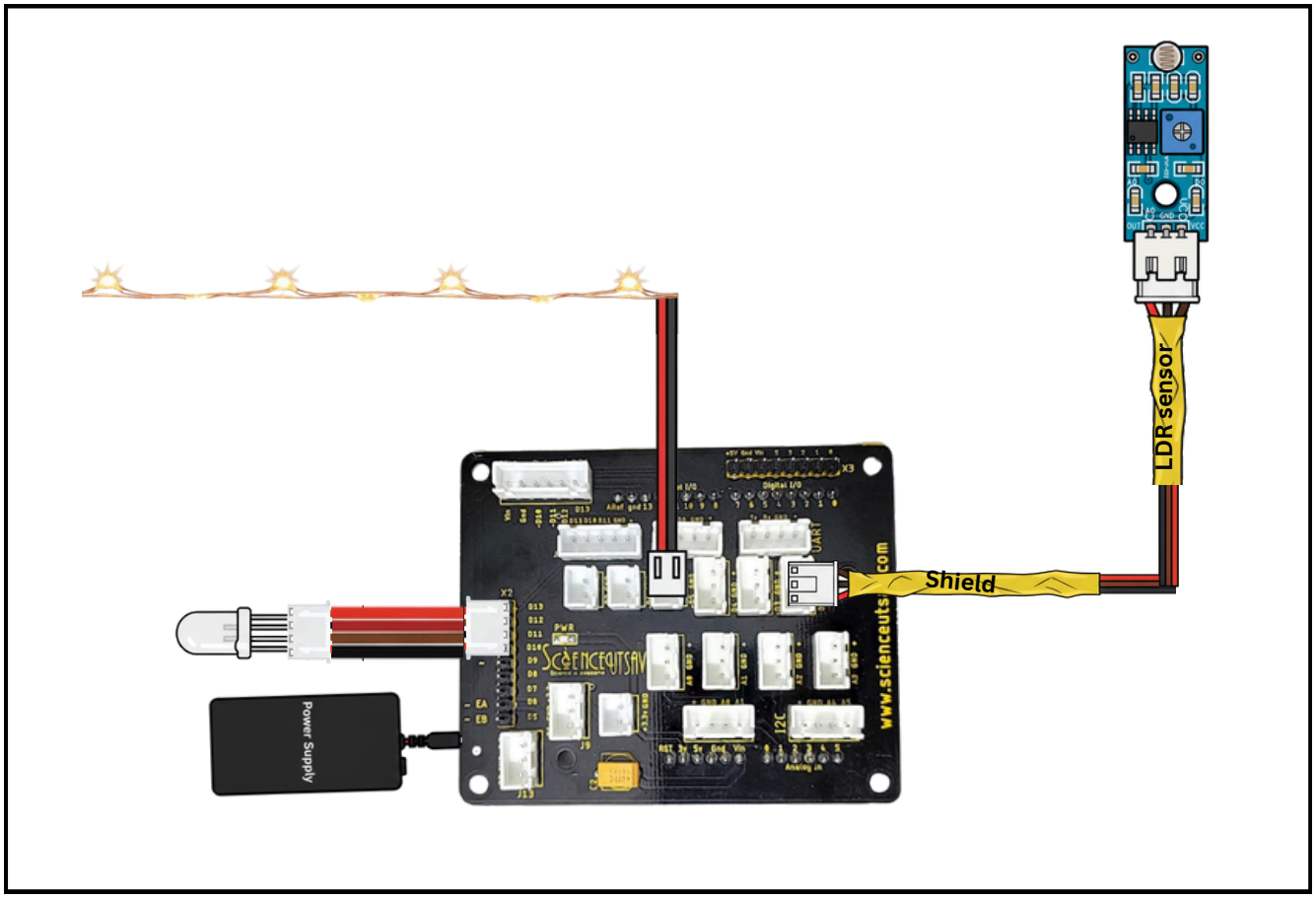
Watch the lamps turn on automatically!

Code

```
/*
ScienceUtsav Arduino Code - Session 6 to 10
Corrected for 3-pin DIGITAL LDR module.
LDR uses digitalRead(), not analogRead().
*/

const int ldrSensor = 2;
const int led1 = 5;
const int led2 = 6;
const int DARK_STATE = HIGH;

void setup() {
  pinMode(ldrSensor, INPUT);
  pinMode(led1, OUTPUT);
  pinMode(led2, OUTPUT);
}

void loop() {
  bool darkDetected = digitalRead(ldrSensor) == DARK_STATE;
  digitalWrite(led1, darkDetected ? HIGH : LOW);
  digitalWrite(led2, darkDetected ? HIGH : LOW);
}
```

EXPERT PROJECT

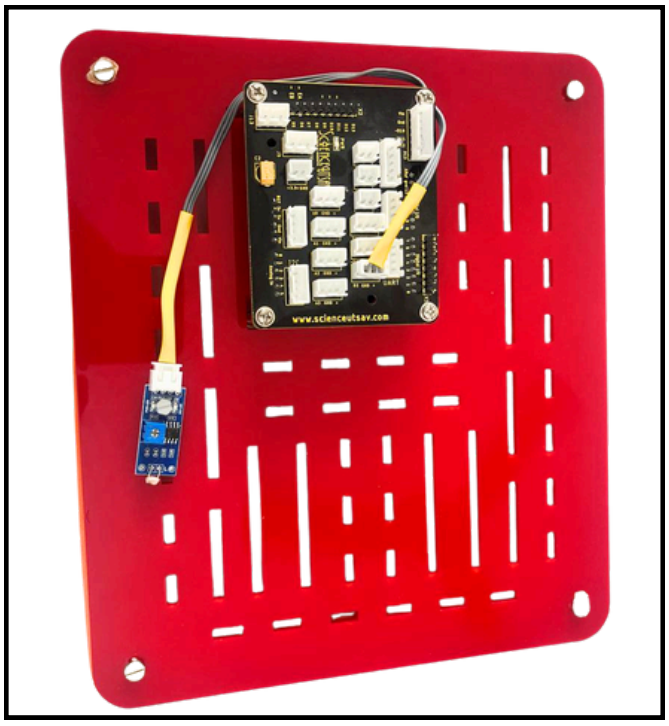
The circuit diagram of the project is given below



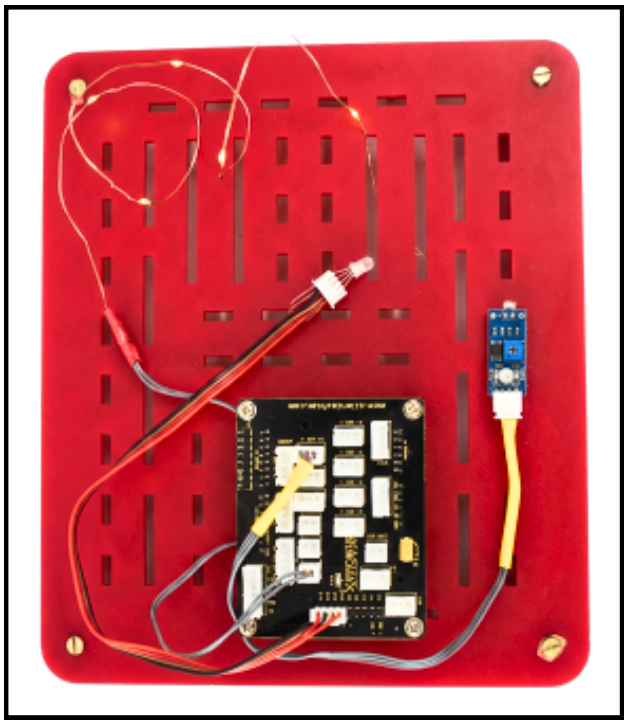
CONNECTIONS:

1. LDR SENSOR - D2
2. STRING LED - D5
3. RGB LED - D10, D11, D13

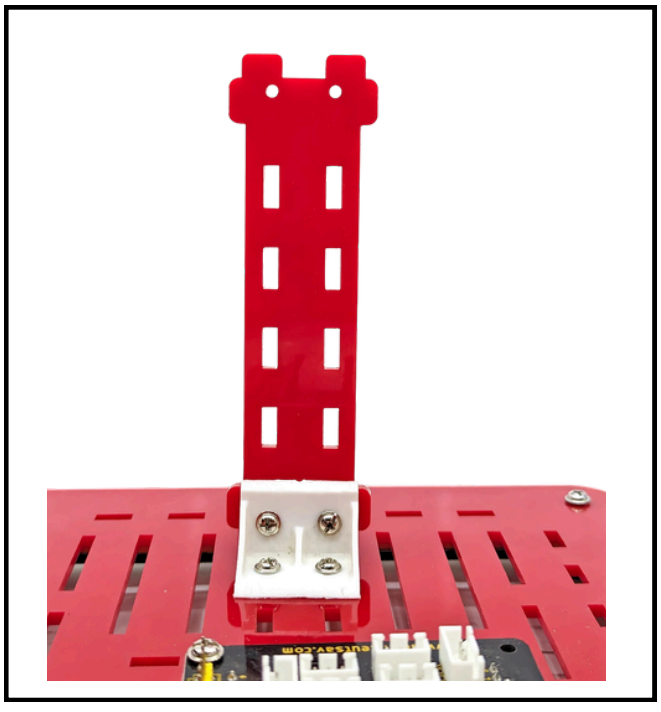
Step 1- Connect the LDR sensor to the board



Step 2- Connect the string LED and the 4-led LED as shown



Step 3- Fix the stand onto the board



Step 4- Attach the template to the stand using bolts

Step 5- String the LED as shown through the holes. Make sure the LED comes to the front

Step 6- Fix the 4-led LED as shown to the template

Get ready to be spooked!

Code

```
/*
ScienceUtsav Arduino Code
Haunted Castle Smooth Light Effect

LDR Sensor = D2
String LED = D5

RGB Common Pin = D12
RGB Red = D13
RGB Green = D11
RGB Blue = D10
*/

const int ldrSensor = 2;
const int stringLed = 5;

const int rgbCommon = 12;
const int redLED = 13;
const int greenLED = 11;
const int blueLED = 10;

void setup() {
  pinMode(ldrSensor, INPUT);

  pinMode(stringLed, OUTPUT);

  pinMode(rgbCommon, OUTPUT);
  pinMode(redLED, OUTPUT);
  pinMode(greenLED, OUTPUT);
  pinMode(blueLED, OUTPUT);

  digitalWrite(rgbCommon, HIGH);  // RGB 4th pin always HIGH
}

void loop() {
  if (digitalRead(ldrSensor) == HIGH) {

    // String LED fade in
    for (int brightness = 0; brightness <= 255; brightness = brightness + 5) {
      analogWrite(stringLed, brightness);

      // RGB red color ON
      digitalWrite(redLED, LOW);
      digitalWrite(greenLED, HIGH);
      digitalWrite(blueLED, HIGH);

      delay(30);
    }
```

```
// String LED fade out
   for (int brightness = 255; brightness >= 0; brightness = brightness - 5) {
     analogWrite(stringLed, brightness);

     // RGB blue color ON
     digitalWrite(redLED, HIGH);
     digitalWrite(greenLED, HIGH);
     digitalWrite(blueLED, LOW);

     delay(30);
   }

   // String LED fade in again
   for (int brightness = 0; brightness <= 255; brightness = brightness + 5) {
     analogWrite(stringLed, brightness);

     // RGB purple color ON
     digitalWrite(redLED, LOW);
     digitalWrite(greenLED, HIGH);
     digitalWrite(blueLED, LOW);

     delay(30);
   }

   // String LED fade out again
   for (int brightness = 255; brightness >= 0; brightness = brightness - 5) {
     analogWrite(stringLed, brightness);

     // RGB green color ON
     digitalWrite(redLED, HIGH);
     digitalWrite(greenLED, LOW);
     digitalWrite(blueLED, HIGH);

     delay(30);
   }

  }
  else {
   // All lights OFF in bright light
   analogWrite(stringLed, 0);

   digitalWrite(redLED, HIGH);
   digitalWrite(greenLED, HIGH);
   digitalWrite(blueLED, HIGH);
  }
}
```